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Assistive Technology and Environmental Control Devices

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SUMMARY PEARLS

- Assistive technology (AT) encompasses devices, systems, and tools that support individuals with disabilities in performing tasks that may otherwise be difficult or impossible.
- AT facilitates recovery and improves patient outcomes by enabling functional independence in daily living activities.
- AT has progressed from simple aids such as crutches in ancient times to modern innovations such as robotic exoskeletons.
- AT development must be personalized, focusing on individual needs, functionality, and adaptability.
- Barriers to AT include cost, complexity, and lack of awareness. Multidisciplinary collaboration and evidence-based practices are essential.
- Integration of artificial intelligence, wearable technologies, and 3D printing will further enhance the effectiveness and accessibility of AT.

Definition of Assistive Technologies

The World Health Organization (WHO) defines assistive technologies (ATs) as “the application of organized knowledge and skills related to assistive products, including systems and services.” AT is a subset of health technology³¹ and encompasses a wide range of devices, systems, and tools designed to support disabled individuals in performing difficult or impossible daily functions. These technologies include simple, low-tech tools such as grab bars and magnifying glasses to sophisticated, high-tech solutions such as powered wheelchairs, communication devices, and exoskeletal robots provided as assistive products. According to the International Organization for Standardization (ISO) definition, assistive product refers to “any product (including devices, equipment, instruments, and software), specially produced or generally available, used by or for persons with disability for participation; to protect, support, train, measure or substitute for body functions/structures and activities; or to prevent impairments, activity limitations or participation restrictions.” The primary goal of ATs is to enhance the independence, productivity, and overall quality of life of individuals facing physical, sensory, or cognitive impairments. “Welfare technology” partially overlaps the concept of ATs, emphasizing the sociotechnologic aspects and integration into health care and social care practices.¹¹

Importance of Assistive Technologies in Rehabilitation

Functional difficulties are a worldwide problem. In a global survey, the prevalence of participants reporting functional difficulties

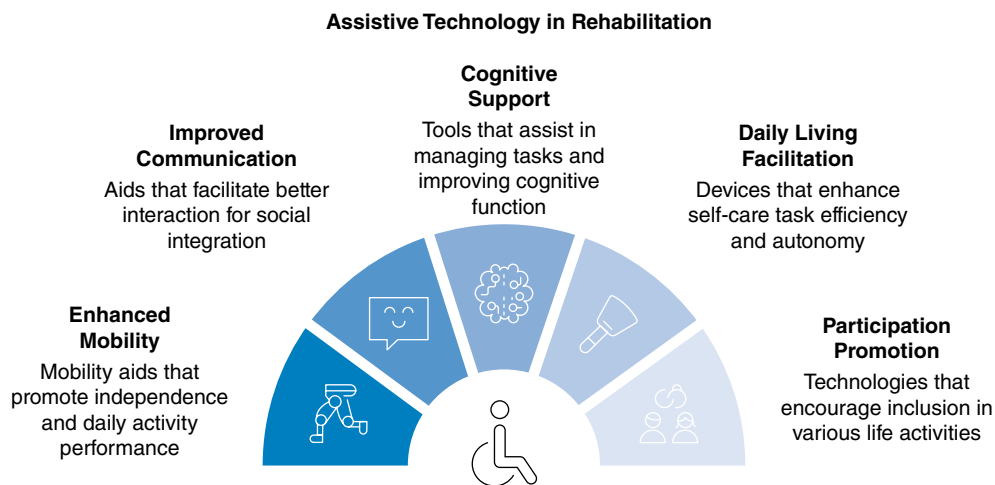
is seeing (20.9%), mobility (12.1%), cognition (6.4%), hearing (4.8%), self-care (4.1%), and communication (2.4%).⁷⁶ ATs can play a critical role in facilitating recovery and improving patient outcomes in both daily living and rehabilitation. The WHO predicts that by 2030 over 2 billion people will need at least one assistive product. Rehabilitation aims to restore or enhance the functional abilities of individuals who have experienced injury, illness, or disability. The demand for ATs is significantly higher in populations undergoing rehabilitation compared to the general population. ATs provide essential support by enabling patients to perform everyday tasks, engage in therapeutic exercises, and communicate effectively, promoting greater participation in various aspects of life. By empowering patients to overcome functional limitations and achieve their personal goals, ATs underscore the importance of innovation and adaptability in the ongoing efforts to improve rehabilitation outcomes. The significance of AT in rehabilitation can be highlighted through several key aspects (Fig. 20.1).

Enhanced Mobility and Independence

Mobility aids such as wheelchairs, prosthetics, and walkers enable individuals to move freely and independently, reducing reliance on caregivers and enhancing the ability to perform daily activities.

Improved Communication

For individuals with speech or hearing impairments, communication aids such as speech-generating devices and hearing aids facilitate



• **Fig. 20.1** Several key aspects in which assistive technology provides significant supports aligning outcomes in rehabilitation.

better interaction with others, improving social integration and emotional well-being.

Support for Cognitive Functioning

Cognitive aids, including memory aids and specialized software, assist individuals with cognitive impairments in managing daily tasks, improving organization, and enhancing overall cognitive functioning.

Facilitation of Daily Living Activities

Devices designed to assist with activities of daily living, such as adaptive utensils and dressing aids, enable individuals to perform self-care tasks more efficiently, fostering a sense of autonomy and self-esteem.

Promotion of Participation

ATs help bridge the gap between individuals with disabilities and their environments, promoting greater participation in educational, vocational, and recreational activities. This inclusion is essential for the holistic rehabilitation of individuals, ensuring they can lead fulfilling and active lives. Rehabilitation professionals should be trained to assess and recommend suitable ATs for each case. Rehabilitation programs should integrate state-of-the-art ATs and remain actively engaged in research to address identified gaps and challenges.

Evolution and Milestones of Assistive Technologies

The history of AT is marked by a continuous evolution driven by human ingenuity and the desire to improve the quality of life for individuals with disabilities. In ancient Egypt, simple wooden crutches were used to aid mobility, while the Greeks and Romans designed early prosthetic limbs made from wood and metal. During the Middle Ages, there were innovations such as hand-operated carts for mobility and magnifying glasses to aid vision. The Industrial Revolution in the 18th and 19th centuries marked a significant turning point in the development of ATs. Engineering and materials science advances have led to the creation of more sophisticated and effective devices. The 20th century saw rapid

advancements in medical technology and rehabilitation methods, significantly impacting the field of AT. Following both world wars, there was a surge in the development of prosthetics and orthotics caused by the increased number of injured veterans. This period also saw the introduction of powered wheelchairs and the initial development of electronic aids for those who were visually and hearing impaired. Table 20.1 summarizes the key milestones in modern AT development.

From ancient rudimentary aids to today's sophisticated digital devices, each milestone has contributed to the empowerment and independence of individuals with disabilities. As technology advances, the potential for further innovations in AT remains vast, promising even greater enhancements in the quality of life and rehabilitation outcomes for those in need.

Principles in Designing and Delivering Assistive Technologies

Designing and delivering ATs require a comprehensive understanding of the users' needs, abilities, and environment. Effective AT should be user centered, functional, and adaptable to ensure it provides meaningful benefits. The framework for developing metrics to evaluate the ergonomic features of ATs has been proposed and summarized.⁷⁴ Several essential frameworks have been proposed that consider the use of scenario and human-technology-environment-social interactions (Table 20.2). The *International Classification of Functioning, Disability, and Health* (ICF) was the most frequently used framework. This framework helps classify the health and disability components of functioning and contextual factors, emphasizing the role of environmental and personal factors in the effectiveness of ATs. The human activity AT is another framework that describes the interaction between human activity, AT, and the contextual environment, highlighting how personal and environmental factors can enhance or restrict the use of AT.⁵

The design and delivery process of ATs should involve an interdisciplinary team, including health care professionals, engineers, therapists, and users, to ensure a holistic approach. Several essential steps in the AT service and delivery process have been recommended: assessment, formulation of a care plan, selection, fitting, training, follow-up, and evaluation (Table 20.3). The AT service delivery

TABLE 20.1 Milestones in Modern Assistive Technology Development

Period	Milestone	Description	Importance
1932	Invention of the modern wheelchair	Harry Jennings invented the first lightweight, collapsible wheelchair	Revolutionized mobility for individuals with disabilities, increasing independence and ease of transportation
1940s-1950s	Development of hearing aids	Transition from bulky, analog hearing aids to smaller, more efficient electronic devices	Making hearing aids more accessible and user friendly with the introduction of transistor technology
1960s-1970s	Introduction of prosthetic innovations	Advances in materials and introduction of myoelectric prosthetics	Greatly improved the functionality and comfort of prosthetic devices
1980s	Rise of computer-assisted technologies	The advent of personal computers and the emergence of software programs aiding communication, learning, and daily living tasks	Provided new opportunities in the digital era of ATs
1990-2000s	Emergence of the Americans with Disabilities Act	Landmark legislation in the United States mandates equal access to public spaces, employment, and communication	Spurred developing and adopting various assistive technologies to comply with accessibility standards
	Integration of assistive technologies (ATs) in education	Emphasis on including ATs in individualized education plans under the Individuals with Disabilities Education Act	Ensured that students with disabilities had access to tools supporting their learning and classroom participation
2010s	Advancements in wearable technology	Development of wearable devices, such as smartwatches and fitness trackers, with integrated health monitoring and assistive features	Made it easier for individuals to manage their health and daily activities. Wearable exoskeletons provide new possibilities for mobility and rehabilitation

TABLE 20.2 Frameworks to Conceptualize the Basis for Measurements in Assistive Technology (AT) Service

Models	Components	Description	Relevance to ATs
International Classification of Functioning, Disability, and Health (World Health Organization) ⁶⁰	1. Body functions and structures 2. Activities and participation 3. Environmental factors 4. Personal factors	A comprehensive framework is used to describe and categorize health and disability. Focuses on human functioning and contextual factors, including environmental influences.	Emphasizes the role of environmental factors in disability and functioning, guiding the development and application of ATs to improve participation and activity performance.
Human Activity AT ¹⁷	1. Human 2. Activity 3. AT 4. Context	Focuses on the relationship between humans and AT within their environment. Aimed at enhancing activity through technologic solutions.	Directly centered on ATs, assessing how devices can support individual activities in specific contexts, promoting optimal alignment between user needs and technologic features.
Canadian Model of Occupational Performance and Enablement ⁷⁰	1. Person 2. Environment 3. Occupation	Highlights the interactive effects of the person, environment, and occupation on one's performance. Includes spiritual, affective, cognitive, and physical components of the person.	Supports the use of ATs by emphasizing personalized, contextually integrated interventions that enable meaningful occupational engagement.
Person-Environment-Occupation-Performance ⁷	1. Person 2. Environment 3. Occupation 4. Performance	A holistic model that examines how personal attributes, environmental factors, and occupations influence an individual's performance.	Useful for identifying how ATs can be tailored to improve performance in meaningful occupations, considering both personal capabilities and environmental barriers or facilitators.

process is attracting interest. A systematic literature review showed that factors in almost all the process steps affect satisfaction with the devices, which can lead to underutilization and abandonment of ATs. A client-centered approach to the process is advocated.⁴² Providing adequate training and support to users and their caregivers is crucial to use ATs effectively. Continuous support and maintenance services should be available to address any issues that arise while using the technology. Finally, the comprehensive process of AT service delivery should be supported by clinical evidence. The main aspects in designing and delivering ATs should be considered (Fig. 20.2).

User-Centered Design

ATs must be tailored to the specific needs and preferences of individual users. This includes considering their physical abilities, cognitive capacities, and personal goals. Users should be actively involved in the design and selection process to ensure the technology meets their needs and fits into their daily lives. The technology should be intuitive and easy to operate, minimizing the learning curve and promoting user independence. Ergonomic principles of ATs have been addressed in specific standards (e.g., ISO 9241-110:2006 or DIN 92419) to enhance the quality of the human technology interaction with ATs.⁷⁴ User feedback and

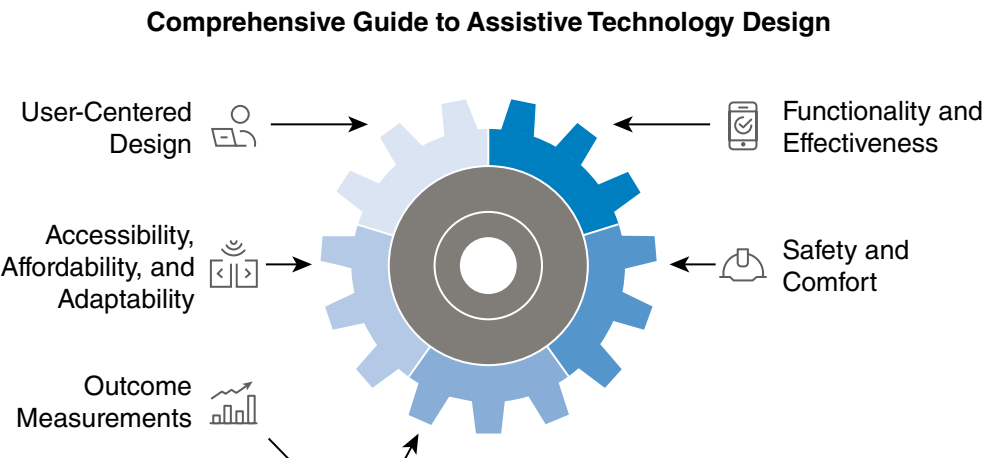
preference are crucial in shaping AT systems to meet users’ real-world needs and preferences. For example, a review demonstrates that individuals who are blind and visually impaired may be willing to sacrifice features for more reliable navigation while considering usability and user experience.⁶⁹ The role of formal and informal caregivers in the context of AT should be noted. Nevertheless, their needs and interactions with technology are often underexplored. Caregivers frequently face the physical and emotional burdens of their work. They use, adapt, and shape ATs in their caregiving roles, pointing to the need for designs that consider the caregivers’ work environments and challenges.⁹

Functionality and Effectiveness

The functionality of the technology in achieving users’ specific goals is related to their mobility, communication, or daily living activities. Human factor and ergonomic considerations are considered in the design and development process of ATs, aiming to emphasize users’ unique requirements.³ The technology must be reliable and durable to withstand regular use and provide consistent support to each user. Regular assessment of the technology’s performance is essential to ensure it continues to meet users’ needs and to make necessary adjustments.

TABLE 20.3 Essential Steps in the Assistive Technology Service and Delivery Process

Step	Objective	Key Elements
Assessment	Identify the individual needs, abilities, and goals.	Needs Analysis, Functional Assessment, Environment Evaluation, Consultation
Formulation of a care plan	Develop a tailored plan to address the identified needs.	Goal Setting, Selection of Interventions, Resource Allocation, Timeline
Selection	Choose the most suitable assistive technology (AT) solutions.	Product, Evidence Based, Reimbursement, Compatibility, Cost-Benefit Analysis
Fitting	Customize and adapt the chosen AT to the user.	Customization, Integration, Adjustment, Initial Feedback
Training	Educate the user on how to use the AT effectively.	Skill Development, Practice Sessions, Support Materials, Ongoing Support
Follow-up	Monitor the user’s progress and address any issues.	Regular Checkins, Problem Solving, Adjustments, Feedback Loop
Evaluation	Assess the overall effectiveness and impact of the AT.	Outcome Measurement, User Satisfaction, Impact Analysis, Reporting



• Fig. 20.2 Main considerations in designing and delivering assistive technology.

Accessibility, Affordability, and Adaptability

ATs should be designed using universal design principles, ensuring accessibility and usability by the widest possible range of users. The technology should be intuitive and easy to use, minimizing the learning curve and reducing the likelihood of user frustration. The AT should be affordable for users, considering their financial situation and potential funding sources, such as insurance or grants. The design should comply with accessibility standards to ensure it is easily used by individuals with various types of disabilities. The technology should be readily available and accessible to obtain, with consideration given to geographic and socioeconomic factors. ATs should be adaptable to the users' changing needs and environmental factors. This includes modular designs that allow for upgrades and modifications as needed. The technology should also be compatible with other devices and systems that users may already be utilizing, thus ensuring seamless integration into their daily routines.

Safety and Comfort

Potential risks and harms associated with using the technology should be thoroughly assessed and minimized. The design should prioritize user comfort and ergonomics to prevent strain or injury from prolonged use. The AT must meet safety standards to avoid accidents or harm to each user. Prosthesis and orthosis can cause skin irritation and pain caused by improper fitting issues. Provision for regular maintenance and access to technical support are crucial to ensure the technology remains safe and functional over time. The safety of ATs is highly dependent on the environment in which they are utilized. For example, wheelchair users may encounter difficulties navigating uneven terrain, leading to accidents. Sideways tipping and forward falls from wheelchairs are highly influenced by curb height and the approach angle. Using seat belts can mitigate the risk of falls and reduce head impact forces during sideways tipping incidents.²⁷ It should be emphasized that some adverse events associated with ATs could be overlooked and are not limited to physical conditions. It has been reported that some technologies may lead to increased anxiety, agitation, or decreased interaction with caregivers, pointing to the need for careful consideration of technology design and its implementation in dementia care.²⁹ It has been proposed that changes in the risk tolerance of children and their parents after the

introduction of ATs or protective equipment for children exist; it is possible AT devices could interact with other known risk factors to promote injury risk-taking behaviors in children.⁵⁷

Outcome Measurements, Impacts, and Evidence-Based Practice

The benefits of using ATs could be multidimensional, including physical, psychological, social, and well-being. Outcome measurements using ATs are fundamental for advancing the field, optimizing individual user experiences, and ensuring efficient use of health care resources. The outcome measurements for ATs are generally categorized into four main types: functional efficacy, satisfaction, psychosocial impacts, and quality of life.¹² Table 20.4 demonstrates some common outcome measurements related to AT service. A system of practice that incorporates a combination of validated tools and measures to address various purposes and perspectives is likely to be beneficial.⁵⁹ Combining these domains in future AT evaluation may provide holistic views of AT utilization.⁶⁸ A relationship has been recognized between the psychosocial impact and the probability of future abandonment of certain ATs.³⁸ The data bridging the gap between clinical expectations and real-world user experiences foster a more responsive and responsible approach to AT deployment. It can improve user satisfaction and quality of life, guide device customization and innovation, enhance clinical decision making, justify funding and insurance coverage, and facilitate policy development and implementation. Several outcome measurements have been developed to demonstrate the reliability, validity, and utility (e.g., Quebec User Evaluation of Satisfaction with Assistive Technology [QUEST 2.0]).⁶ Nevertheless, using these measurements should still be reverified for different social and cultural backgrounds. Some areas, such as mobility, have numerous studies, whereas other domains, such as ATs for domestic activities and communication, are less studied.¹²

Using adequate ATs can also improve clinical outcomes for specific populations. For example, a study reported that patients with personal hearing amplifiers in emergency departments and acute care settings showed improved engagement of medical instructions and were less likely to revisit the emergency department within 30 days.¹³ Access to suitable wheelchairs with proper cushions significantly reduces the frequency of pressure sores or urinary tract infections.¹⁹ On the other hand, discontinuation of

TABLE 20.4 Common Outcome Measurements Related to Assistive Technology Service

Domains	Measurements Examples	Description
Functional Efficacy	Functional Independence Measure ⁴⁵	Evaluates the general level of a patient's disability and how much assistance the patient requires
	Wheelchair Skills Test ⁴⁰	Assesses wheelchair users' ability to perform specific maneuvers
Satisfaction	Quebec User Evaluation of Satisfaction (QUEST 2.0) ²³	Measures user satisfaction with the assistive device and associated services
	Wheelchair Outcome Measure ⁵¹	Evaluates satisfaction in terms of how well the wheelchair meets users' needs
Psychosocial Impact	Psychosocial Impact of Assistive Devices Scale ²²	Measures changes in self-esteem, adaptability, and competence caused by assistive technologies
	Communicative Effectiveness Index ⁴⁷	Evaluates the impact of communication devices on speech efficiency
Quality of Life	36-item Short Form Health Survey (SF-36) ⁷³	A generic health-related quality of life measure for physical and mental health
	World Health Organization Quality of Life ²⁴	Assesses detailed quality of life, including physical, psychological, and social

assistive device use was associated with an increased risk for severe falls in older adults living at home, which has been reported.⁵⁸

ATs significantly enhance educational engagement and psychosocial well-being for students with various disabilities within higher education institutions.⁵⁰ When effectively implemented in work environments, ATs can improve employee productivity.⁶² Therefore ATs may reduce the risk of joblessness and poverty often experienced by people with functional difficulties. The provision of ATs involves a thoughtful and comprehensive approach that prioritizes users’ needs and preferences. By adhering to these principles, designers and health care providers can ensure that ATs are effective, accessible, and beneficial to those who rely on them. This user-centered focus not only enhances the quality of life for people with disabilities but also promotes greater inclusion and participation in all aspects of society. Because of the high heterogeneity of disabilities and the difficulty of conducting sham control trials in the AT field, high-quality evidence to determine AT effectiveness remains relatively scanty. In addition, more comparative and longitudinal studies of ATs and research involving diverse groups within and across countries should be emphasized. Therefore enhancing data collection methods in the real-world setting by integrating more specific questions about AT into existing health and disability surveys will improve the quality and comparability of data across studies. Updating and maintaining large international data repositories like the AT2030 data portal is vital to support research and policymaking by providing a source of up-to-date information on AT access.⁸ This allows stakeholders, including researchers, policymakers, and advocacy groups, to identify gaps in AT provision.²¹

Classifications of Assistive Technologies

The AT classification system is complex and can be categorized based on functionality, type of assistance provided, or specific needs they address. The classification systems for ATs are summarized in Table 20.5. The WHO provides the ICF classification system, which is a framework that classifies AT based on how it supports participation and activity, considering factors such as body functions and structures, activities and participation, and environmental factors. Following is an overview of the main taxonomy and classifications within AT.

Mobility Aids

Mobility aids are designed to assist individuals with mobility impairments to move around or manipulate objects more efficiently and safely. Sophisticated devices are designed according to specific needs. Modern wearable robotic suits assist individuals with severe mobility impairments in walking. Exoskeletons have shown promise in both rehabilitation settings and for daily use, enabling users to stand and walk with greater stability. Modern wheelchairs are equipped with intelligent technology, such as GPS, collision detection, and self-balancing systems, which enhance users’ mobility and safety. Mobile arm support is designed to assist with upper extremity motor skills, compensating for muscle weakness to perform diverse activities such as opening a container and handwriting. Examples include the following:

Walking aids: Canes, quad canes, and crutches provide basic support and balance for walking.

TABLE 20.5
Common Classification Systems for Assistive Technologies (ATs)

Frameworks and Organizations	Description	Examples of Classification	Classifications and Categories
International Organization for Standardization 9999:2022 International Organization for Standardization	An international standard that classifies assistive products for persons with disabilities. Under the classification of assistive products covers >900 types.	Mobility aids, communication devices	12 one-level classifications: measuring, stimulating, or training physiological and psychological functions; orthoses and prostheses; self-care activities and participation in self-care; relating to personal mobility and transportation; activities and participation in domestic life; furnishings, fixtures, and other assistive products for supporting activities in indoor and outdoor human-made environments; communication and information management; controlling, carrying, moving, and handling objects and devices; controlling, adapting, or measuring elements of physical environments; work activities and participation in employment; recreation and leisure.
World Health Organization's Global Cooperation on Assistive Technology World Health Organization	Aims to improve access to assistive technology globally, focusing on essential aspects of ATs.	Wheelchairs, hearing aids, visual aids	Based on the Priority Assistive Products List (APL), a core set of ATs includes hearing aids, wheelchairs, communication aids, spectacles, artificial limbs, pill organizers, and memory aids. Additionally, a broader functional classification used in related contexts.
Rehabilitation Engineering and Assistive Technology Society of North America	Focuses on standards, development, and education in AT.	Communication aids, environmental controls	Varies by specific area: various specific standards.
Healthcare Common Procedure Coding System Centers for Medicare and Medicaid Services	Includes codes for billing and insurance purposes in the United States, covering various medical equipment.	Prosthetics, orthotic devices, durable medical equipment	Thousands of codes: includes various medical and assistive devices. Key codes include assistive technology assessment, therapeutic services for speech-generating devices, assistive hearing devices and related supplies, prosthetic implants and auditory devices, mobility devices like transport chairs, and remote patient monitoring devices and services.

▶ **Wheelchairs:** Manual and powered wheelchairs (Video 20.1) provide mobility for individuals who cannot walk.

Scooters: Electric scooters offer an alternative to wheelchairs for those with limited walking ability.

Walkers and rollators: These provide support and stability for individuals who can walk but need assistance. They are equipped with wheels and often come with seats and brakes, allowing easier movement over smoother surfaces. Some power wheelchairs can even overcome obstacles and environmental barriers, such as step-climbing power wheelchairs.⁶⁷

Orthoses: Orthoses are devices used to support, align, prevent, or correct deformities or to improve the body's function. They can be custom made or prefabricated and designed to fit specific body parts, such as the limbs or trunk (Fig. 20.3).

Prosthetics: Artificial limbs that replace missing body parts, enhancing mobility and function.

Exoskeletons and robotic devices: Exoskeletal devices work in conjunction with the user's body. Depending on the design and intended use, they can be powered by motors and actuators or purely mechanical (passive) (Fig. 20.4).

Communication Aids

Communication aids, or augmentative and alternative communication (AAC) devices, help individuals with speech or language impairments communicate more effectively. Modern AAC devices incorporate advanced features such as predictive text, voice synthesis, and customizable interfaces. These devices are also becoming more portable and affordable. Improved algorithms and machine learning techniques have made speech recognition software more accurate and responsive, allowing users to control devices and communicate more effectively. Examples include the following:

Speech-generating devices: Electronic devices that produce spoken words when the user selects symbols or types of text.

Communication boards: Boards with pictures, symbols, or words that individuals can point to in order to communicate. Programs or apps functionalized with iPad, iPhone, Windows, or Android systems are available (Fig. 20.5).

Text-to-speech software: Software that converts written text into spoken words, useful for individuals with speech impairments.



• **Fig. 20.3** A dynamic hand orthosis that trains patients' hand function by assisting finger extension. (From iOpenDynamic, BALDUR Medical, Taiwan.)

Cognitive Aids

Cognitive aids are tools designed to support individuals with cognitive impairments in managing daily tasks, improving cognitive function, and compensating for difficulties in memory, attention, problem solving, and other cognitive functions (Fig. 20.6). Examples include the following:

Memory aids: Devices or apps that help individuals remember important tasks, appointments, or information (e.g., electronic organizers, reminder apps).

Specialized software: Programs to enhance cognitive skills such as attention, memory, and problem solving (e.g., brain training apps, educational software).

Electronic calendars: Digital calendars can be set with reminders and alerts to help manage schedules and tasks.

Sensory Aids

Sensory aids assist individuals with sensory impairments, such as vision or hearing loss, by enhancing their ability to perceive their environment. Advanced hearing aids have Bluetooth connectivity, noise cancellation, and directional microphones, providing a better listening experience. Some models also integrate with smartphones, allowing for remote adjustments and settings customization. Haptic ATs utilize touch and tactile feedback to assist individuals with hearing or visual impairments. These technologies convert auditory or visual information into tactile sensations, enabling users to perceive and interact with their environment more effectively.⁶³ Technologic advancements have led to the development of sophisticated vision aids, such as smart glasses that use augmented reality to enhance visual perception and electronic implants that can partially restore vision for individuals with certain types of blindness. Examples include the following:

Hearing aids and personal hearing amplifiers (PHAs): Devices that amplify sound for individuals with hearing loss. Hearing aids are custom-fit devices to compensate for specific hearing deficits. PHAs are generally over-the-counter products that amplify sounds to make them louder and clearer.

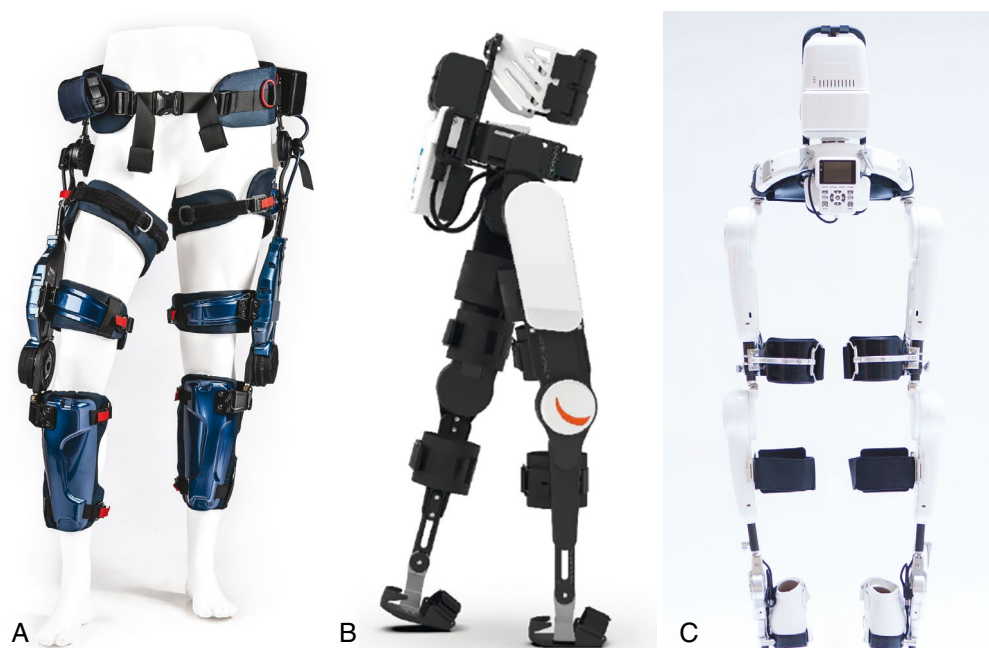
Cochlear implants: Surgically implanted devices that provide a sense of sound to individuals with severe hearing loss.

Vision aids: Tools such as spectacles, magnifiers, screen readers, and braille displays that assist individuals with visual impairments depending on their severity and types (Video 20.2).

Navigation systems: Systems that are tailored for individuals with visual impairment, spanning traditional tools such as the white cane to modern technologic solutions such as electronic devices and software applications. These solutions are categorized into visual imagery, nonvisual data, map-based systems, and 3D sound systems. These technologies contribute to obstacle detection and route optimization and provide real-time feedback, thereby improving independence and safety for users.²

Alerting devices: Devices that use visual or tactile alerts to signal important events, such as doorbells, alarms, or phone calls, for individuals with hearing loss.

Medications assistive devices: Individuals with sensory impairment could have an increased risk of medication-related iatrogenic disease. Specialized ATs could support medication management for individuals with sensory impairment to reduce harm. The range of technologies and devices includes applications for Android phones, audio-prescription labeling/reading systems, touch-to-speech devices, and magnifying technology.¹⁸



• **Fig. 20.4** Exoskeletal robots for lower limbs. (A) Keeogo (B-Temia) (Wistron Medical Technology), triggered by movement sensors and algorithms, adapts to the user's movements in real time, providing the necessary support. (B) Free Walk (FREE Bionics Taiwan Inc.), robot assistance to stand, walk, and perform physical training, operated with a wireless control. (C) HAL (CYBERDYNE, Inc.), controlling the exoskeleton through the detected bio-electrical signals of users' intention, providing the necessary support to execute the movement. (A, Courtesy Wistron Medical Technology; B, courtesy FREE Bionics Taiwan Inc.; C, courtesy CYBERDYNE, Inc.)



• **Fig. 20.5** ACC How-Go-Tone is an iOS-based app that provides a customized user interface and allows patients to communicate by touching screens or using specialized triggers or pointing devices. (From Assistive Technology Engineering Lab, Inc.)

Daily Living Aids

Daily living aids are tools designed to help individuals perform everyday activities more efficiently and independently. Examples include the following:

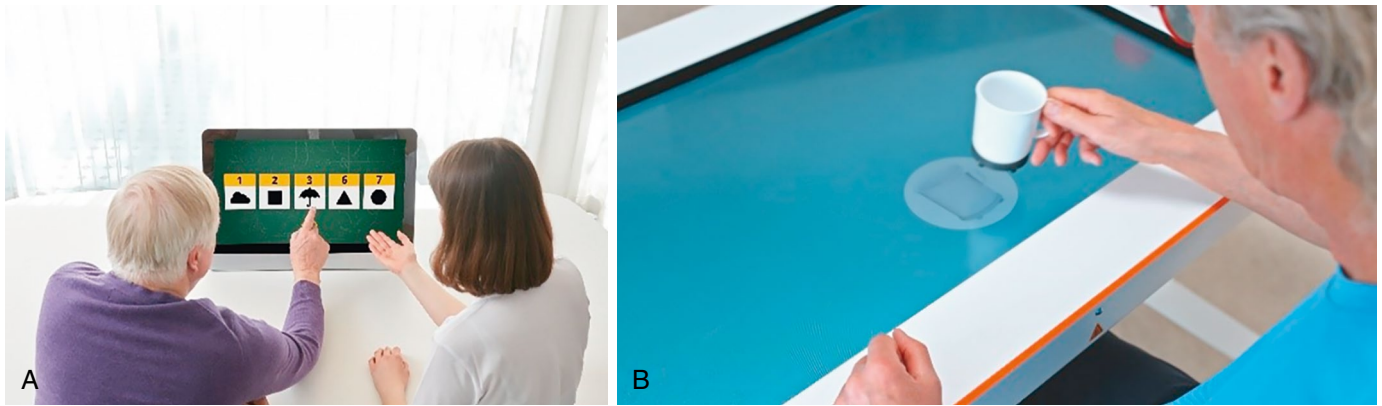
Adaptive utensils: Modified eating utensils with ergonomic handles or specialized grips to assist individuals with limited hand function.

Dressing aids: Tools such as button hooks, zipper pulls, and dressing sticks that help individuals dress independently.

Bathing aids: Devices such as grab bars, toilet lift seats, shower chairs, shower stools, and bath lifts enhance safety and independence in the bathroom.

Environmental Control Systems and Home Automation

Home automation refers to installing technology that automates or remotely controls household functions via devices and programs linked with a wired or wireless connection. Integrating Internet of



• **Fig. 20.6** Cognitive training devices. Neofect Cognitive training (A) allows for practice of multiple executive functioning domains. Users receive performance feedback, promoting skill improvement. MYRO® (B) enables task-oriented rehabilitation with real objects and trains the patient's cognitive abilities. (A, Courtesy Neofect Cognition; B, Courtesy Tyromotion, Inc.)

Things devices into assistive devices allows for seamless control of home environments. Intelligent lights, thermostats, and appliances can be controlled via voice commands, mobile apps, or adaptive switches, improving accessibility and convenience. Modern environment control systems are built on various wireless systems, each offering unique ways to manage and control a user's environment. Infrared control systems, Bluetooth systems, and Wi-Fi are popular choices for home automation. These wireless systems are also helpful for individuals who require remote access to their devices. Except for wireless systems, the advancement of user interface systems makes environmental control user friendly. Devices such as Amazon Echo and Google Home have become central hubs for AT, enabling users to perform various tasks through voice commands, from setting reminders to controlling household devices. This technology can control various aspects of a home environment, including doors, blinds, heating and cooling systems, lighting, windows, doorbells, intercom systems, taps, showers, and entertainment systems through smartphones, tablets, and voice-controlled systems. The primary aim of home automation is to increase independence and enhance the quality of life for individuals, particularly those with physical, sensory, or intellectual disabilities.¹⁵ Home automation also facilitates greater social interaction and community engagement by enabling individuals to communicate and participate in social activities more easily. Improved self-esteem, reduced anxiety, and better psychological well-being among users have been shown in disabled patients using home automation to enhance independence and autonomy.¹⁵ The participants experience reduced social isolation and increased interaction with friends and family through smart devices for communication.¹⁵ Examples include the following:

- Voice-controlled devices:** Smart home devices such as Amazon Echo or Google Home that can control lights, thermostats, and other appliances through voice commands.
- Switch-activated controls:** Systems allow users to operate devices using simple switches that can be activated by various body movements.
- Remote controls:** Devices that enable users to control electronic appliances and systems from a distance.

Computer Access Aids

Computer access aids are tools and devices that enable individuals with disabilities to use computers and mobile devices more

effectively. These aids allow users to perform tasks such as typing, navigating the internet, and operating software applications. Examples include the following:

- Alternative keyboards:** Keyboards with different layouts, larger keys, or customizable settings to accommodate various physical needs.
- Trackballs and joysticks:** Pointing devices that offer alternative methods of controlling the cursor on a screen.
- Eye-tracking device:** A tool for controlling a computer or other devices using eye movements. This device is especially beneficial for people with limited mobility or severe physical disabilities. The device typically includes a camera that tracks the user's eye movements, translating them into commands.
- Screen readers:** Software reads aloud the content displayed on a screen, useful for individuals with visual impairments.
- Speech recognition software:** Programs convert spoken words into text, allowing users to control computers and write documents using their voice.
- Computer access software:** Programs that allow users to directly access other accessibility/usability features on computers, such as text size, magnifier, and read selection.

Educational Aids

Educational aids are tools designed to support learning and academic performance for individuals with disabilities. These aids can be physical devices, digital tools, or instructional materials that help facilitate the acquisition of knowledge and skills. They are particularly beneficial in providing personalized learning experiences, accommodating diverse learning styles, and addressing specific educational needs. Examples include the following:

- Adaptive learning software:** Programs that adjust the difficulty level based on the user's performance and needs.
- Audiobooks and ebooks:** Accessible formats of books that can be read aloud by a computer or other device.
- Specialized educational apps:** These applications support various learning styles and needs, such as visual aids, interactive activities, and customized exercises.
- Speech-to-text technology:** Software or system that converts spoken language into written text. This technology is useful for creating documents, sending messages, or performing any other activity that typically requires typing.

Caregiver Aids

ATs, such as grab bars, sensor systems, and companion robots, are examined for their potential to alleviate some physical and psychological stresses on caregivers. For example, sensor and monitoring systems can enhance the safety and coordination of care in care homes by providing alerts and monitoring residents' activities, thus reducing constant physical supervision needs and allowing caregivers to manage their tasks more effectively.¹⁴ The slide boards used by home care aides to assist client transfers highlight proper equipment that can significantly reduce biomechanical loading on caregivers, thereby potentially reducing injuries and physical strain.¹⁴ Examples include the following:

Transfer assistive devices: These devices, such as sliding boards, lifters, raisers, and supporting belts, help caregivers transfer patients effectively and safely (Fig. 20.7).

Real-time locating systems: These systems automatically identify and track the location of objects or people in real time, typically within a building or other contained area. They can potentially improve residents' safety, quality of care, and efficiency of long-term care facilities.³³

Assistive Technology Needs and Considerations for Specific Populations

Older Adults

Aging is a complicated, dynamic, and individualized process. Accelerating population aging is a global issue. How to achieve successful and healthy aging and the rising demand for aging-in-place technologies have drawn increasing attention.⁵⁵ Living independently can be challenging for seniors. Furthermore, the global challenge of an aging population outpacing the availability of caregivers necessitates technologies that enable older adults to live independently at home.⁶⁴ Cochrane reviews demonstrated that home fall-hazard interventions, including ATs, could reduce the overall rate of falls by 26%; however, these interventions were more effective in people with a higher risk of falling.¹⁶ Among people who fall at home, seniors without using walking aids could sustain more severe injuries.⁴⁸ On the other hand, the prevalence of chronic diseases is significantly higher in the geriatric population compared to the

general population. It is noteworthy that aging people with chronic diseases may also require technologies tailored to managing specific symptoms or conditions, adding to the complexity. For instance, those with diabetes might benefit from devices that monitor blood glucose levels and alert users to take insulin. Walking aids may reduce functional decline in sarcopenic patients.⁶¹ Given that these chronic conditions can come with physical, sensory, or cognitive impairments, the user interface of ATs needs to be particularly intuitive and easy to use for those with specific disabilities (Fig. 20.8). Furthermore, there is often a higher need for ATs to integrate seamlessly with other health care systems to ensure continuous care. This includes the ability to share data with health care providers easily. Technologies for the general older population may focus more on enhancing daily living than integrating closely with medical care.¹⁰

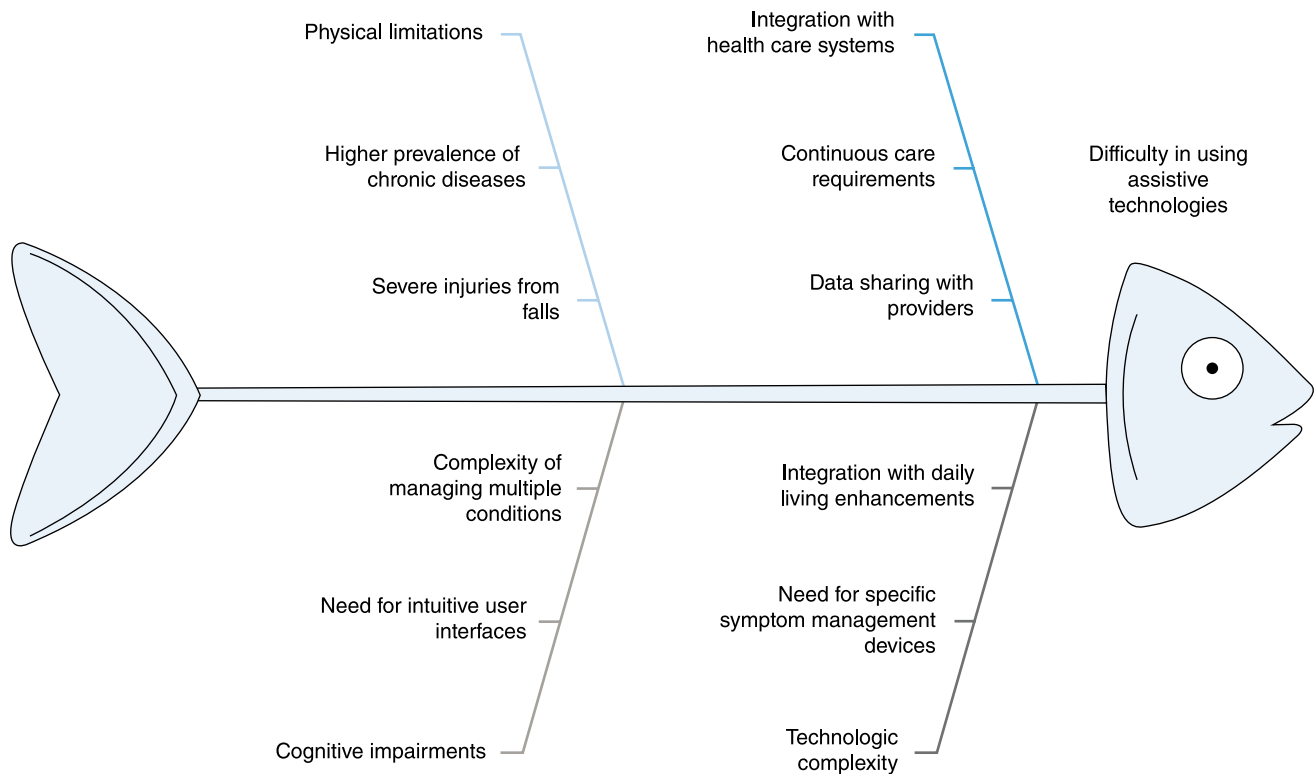
Children

Pediatric populations using ATs have specific characteristics to consider when designing, prescribing, and managing ATs.²⁸ Children grow rapidly, so their assistive devices may need frequent adjustments, resizing, and updates to accommodate their physical growth and changes in their abilities and needs. ATs must support and not hinder the achievement of developmental milestones, such as motor skills, social skills, and language development. Furthermore, it has been noted that the widespread lack of ATs for children with disabilities leads to lower primary school completion rates. Education needs should be considered explicitly in children with disability.⁵³ Incorporating ATs promotes educational engagement and learning efficacy, which is paramount in care. Devices must be user friendly and suitable for children, often incorporating engaging and appealing elements to younger users. This can include using bright colors, interactive features, and designs that appeal to their imagination and interests. Family members and caregivers play a critical role in successfully utilizing ATs. They also need proper training and support to integrate these technologies into a child's daily routine. Emotional and practical support for families is essential because managing assistive devices can add to the caregiving burden.⁵⁴ The family-centered model could be helpful for AT provision for the pediatric population.⁷¹



• **Fig. 20.7** (A, B) Standing-and-raising aids. SARA Combilizer is an aid that helps patients perform sit-to-stand, transfer, and mobilize out of bed easily and safely. (Courtesy Arjo)

Challenges in Implementing Assistive Technology for the Geriatric Population



• **Fig. 20.8** Challenges in implementing assistive technology for the senior population.

Future Directions of Assistive Technology

The field of ATs has seen significant advancements in recent years, driven by rapid developments in engineering, materials science, and digital technology.¹ These innovations have greatly expanded the capabilities of assistive devices, making them more effective, accessible, and user friendly. As technologies continue to evolve, the future of AT holds exciting possibilities that promise to enhance the lives of individuals with disabilities further. This section highlights key areas of development.

Increased Personalization and Customization

Three-dimensional scanning helps with personalized customization and optimizes comfort and functionality for individual users. Three-dimensional printing technology will create highly customized assistive devices that fit the user's unique anatomy and needs.⁷² This can include custom prosthetics, orthotics, and adaptive tools. Machine learning algorithms that adapt in real time to the user's behavior and preferences will make assistive devices more intuitive and effective. Researchers are exploring the use of intelligent materials that can adapt to environmental changes, providing more responsive and adaptable solutions for users.^{39,44}

Integration of New Sensors With Wearable Technologies and Health Data in Assistive Technologies

Advanced sensors have evolved to be multifunctional, highly integrated, and compact.⁴⁴ They can be mounted on clothing,

devices, or bodies to continuously monitor physiologic signals and enhance interaction and communication with the environment and equipment.^{32,39} Wearable devices increasingly incorporate health monitoring features, such as tracking vital signs, detecting falls, and alerting caregivers in case of emergencies. This integration will provide users with continuous health monitoring and proactive health care management. Wearable technology can also help track physical activity, provide feedback, support rehabilitation efforts, and promote an active lifestyle. Artificial intelligence (AI)-powered applications are increasingly used to create personalized learning experiences and cognitive training programs. These tools adapt to the user's progress and provide tailored exercises to improve cognitive functions. Wearable devices and smartphone apps that use AI help individuals with memory impairments by giving reminders, tracking daily activities, and organizing schedules.

Advanced Robotics and New Artificial Intelligence Models

Robots capable of performing a wide range of tasks, from fetching items to providing companionship, will become more common. These robots will significantly benefit individuals with severe mobility impairments or cognitive disabilities.⁴ AI-powered virtual companions will offer emotional support, cognitive stimulation, and assistance with daily tasks, enhancing users' overall quality of life. Generative AI (GAI) refers to a subset of AI technologies capable of generating new content that resembles existing data. This can include text, images, audio, and other forms of media or data. The key attribute of GAI is its ability to learn from a set of data and then use that learned information to create new data instances.

GAI enhances existing components of the brain-computer interface to enhance user interface and experience and to enable the development of new, more integrated applications that can convert brain signals directly into outputs such as speech, text, or images.²⁶

Virtual Reality in Assistive Technologies

Virtual reality (VR) is a computer-generated simulation of an environment that allows users to interact with it in a seemingly natural or physical way through special electronic equipment, such as headsets with screens or gloves fitted with sensors. VR can be categorized into different types, from nonimmersive VR to augmented reality and mixed reality. VR becomes one of the valuable mediums for delivering therapy by creating an immersive environment that allows users to engage in scenarios that feel real, which is particularly beneficial for therapeutic and educational applications. VR environments are highly interactive, safe, and well controlled. VR systems can be extensively customized to meet the specific needs of individuals with different disabilities. This adaptability ensures that VR can assist a wide range of disabilities, from motor and cognitive impairments to sensory and communication disorders.^{41,65} The users can practice and learn new skills without the real-world consequences of failure (Fig. 20.9). This is particularly important for users with severe impairments or those recovering from trauma or disease.

Barriers and Challenges in Using Assistive Technologies

The underadoption of ATs has been recognized in real-world circumstances. A survey of the adoption rates that included 22,666

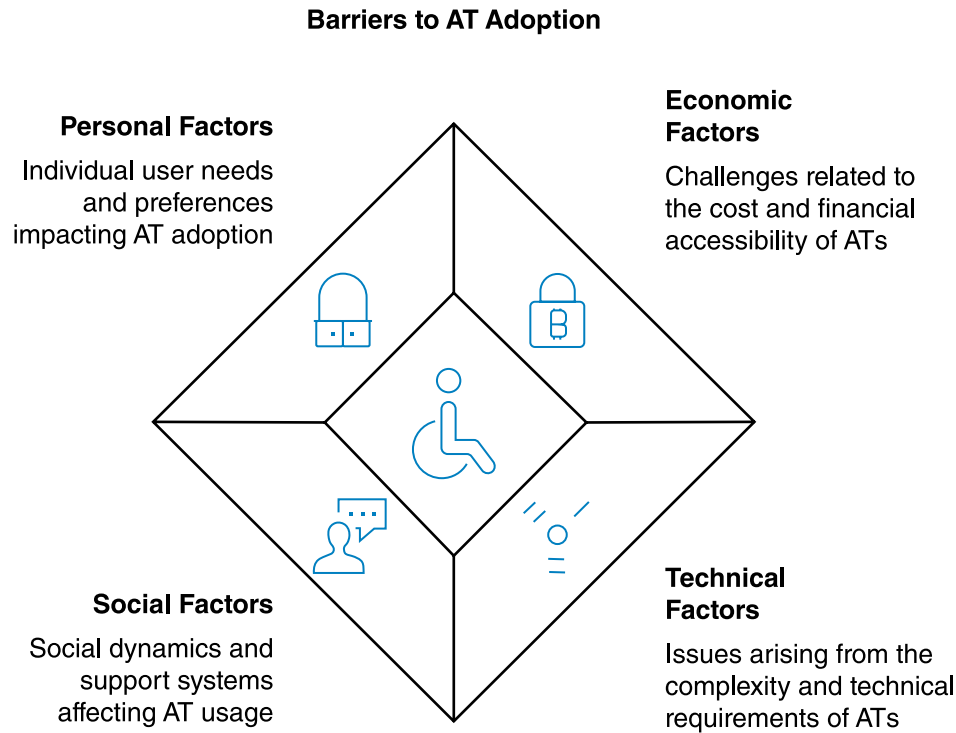
care-dependent persons and caregivers reports that sophisticated types of ATs are still rather scarcely used for home care arrangements.³⁷ Although ATs offer significant benefits to individuals with disabilities, several barriers and challenges can hinder their practical use. These obstacles can stem from various sources, including economic, technical, social, and personal factors. There are multiple barriers across different domains, such as cost, complexity of interventions, and lack of evidence of effectiveness (Fig. 20.10). For instance, product-related factors such as comfort, durability, fit, and limited choices were significant barriers.⁵ This discrepancy between users' needs and the system's provision can result in abandonment, especially when the users need to adapt to the technology (technology centered).³⁶ Understanding these challenges is crucial for developing strategies to overcome them and ensure that ATs can reach their full potential in improving users' lives.⁴⁹ Many patients face significant delays or outright denials when receiving these technologies. This lack of timely access can adversely impact their recovery. The study identified that only 55% of the assistive devices were delivered to children, with delivery times ranging from 12 to 250 days, showcasing considerable variability and potential for delayed rehabilitation.²⁸ Another example is a gap in the availability of modern braille technology, such as refreshable braille displays, especially in less affluent regions or schools with limited funding. This disparity limits students' exposure to braille, affecting their literacy and overall academic performance.

Economic Barriers

Over 2.5 billion people currently require one or more assistive products, which is expected to rise to 3.5 billion by 2050. Access



• **Fig. 20.9** (A, B) HappyGoGo (LongGood Inc, Taiwan) is a virtual reality/augmented reality (VR/AR) device with a depth camera that provides gamification of movement training. (C, D) A game-based VR/AR system for physical and mental health developed by Taipei Medical University.



• **Fig. 20.10** Barriers and challenges in using assistive technology (AT).

disparities are significant, with only about 3% of those in low- and middle-income countries having access to the ATs they need, compared to up to 90% in high-income countries.⁴³ Many assistive devices are expensive, making them inaccessible to individuals with limited financial resources. The high development, customization, and maintenance costs can also be prohibitive. Insufficient funding and reimbursement options from insurance companies and government programs can limit users' ability to afford necessary ATs.

Technical Barriers

Some ATs are complex and difficult to use, especially for individuals with limited technical skills. Poor usability can deter users from adopting these devices. Adequate user interface design, known as human-technology interface (HTI), should be tailored to target users. HTI consists of different kinds of interaction, including brain-computer interface, human-robot interaction, and speech-, gesture-, tongue-, eye-, and multimodal-based interactions.⁷⁴ As more ATs and systems are introduced and launched, interoperability between them should be carefully evaluated. Assistive devices may not always be compatible with other technologies or systems, leading to difficulties in integration and functionality. The need for regular maintenance and the unavailability of repair services can be significant challenges, especially in remote or underserved areas.

Social and Cultural Barriers

The use of ATs can sometimes carry a social stigma, leading to reluctance or embarrassment in using these devices publicly. This can affect the user's willingness to adopt and consistently use ATs.

Cultural attitudes toward disability and technology can influence the acceptance and use of assistive devices. In some cultures, there may be resistance to adopting new technologies. Potential users, caregivers, and health care providers often lack awareness of the available ATs and their benefits, particularly in this field, which advances and evolves quickly. Technology should empower users by enhancing their capabilities and promoting independence rather than creating dependence or stigmatization. Strategies to increase technology acceptance among the population might include targeted education to improve digital literacy and designing user-friendly technology solutions. Insufficient training for users and their support networks can result in improper use or underutilization of assistive devices. Training is essential for maximizing the effectiveness of these technologies. The integration of AT provision training within the curricula of various professional rehabilitation organizations should be highlighted. There is an emphasis on the need for educational programs to include and expand upon this area to ensure entry-level practitioners are competent. Another example is a notable deficiency in braille-trained teachers and the inclusion of braille in the curriculum. This lack of expertise leads to insufficient support for braille as a medium of instruction, impacting students' ability to learn effectively.

Environmental and Contextual Barriers

The physical environment, including inaccessible buildings, transportation, and public spaces, can limit the effectiveness and use of ATs. Limited access to the internet and digital services in some regions can prevent individuals from utilizing modern ATs that rely on connectivity and digital platforms. Workplace accommodation is an important factor in supporting individuals with disabilities in using specialized equipment at work.⁶⁶

Policy and Regulatory Barriers

Developers and manufacturers must navigate the complexities of intellectual property law to protect their innovations while respecting the rights of others. On the other hand, consideration should be given to the benefits of open-source technologies and shared innovation to promote wider access and collaboration. Lengthy and complex regulatory approval processes can delay the availability of new or innovative ATs. Navigating these regulations can also be challenging for developers and manufacturers. Furthermore, inconsistent standards and regulations across different regions can create barriers to the distribution and use of ATs. Inadequate policies and government support can limit access to ATs. Policies are needed to ensure funding, support services, and incentives for developing and distributing these devices.

Personal Barriers

Individual preferences and comfort levels with technology can vary significantly, affecting the adoption and consistent use of assistive devices. The outlook and visibility of assistive devices also affect the acceptance and self-agency of people with disabilities.⁴⁶ Some individuals may have physical or cognitive limitations that make it difficult to use certain types of ATs effectively. Resistance to change and adaptation to new technologies can be a significant barrier.²⁰ Users may feel overwhelmed or intimidated by the introduction of new devices.⁷⁵ Worth noting, studies showed that participants with substantial or severe fragility showed no effectiveness in improving quality of life by ATs addressing age-related difficulties.³⁰ Clinicians should be aware of the subgroup during the provision of ATs. The barriers and challenges in using ATs are multifaceted and interrelated. Addressing these obstacles requires a comprehensive approach that includes improving affordability, enhancing usability, raising awareness, and fostering supportive policies. By tackling these challenges, stakeholders can work toward making AT more accessible and effective for individuals with disabilities.

Ethical and Legal Considerations of Assistive Technologies

The use of ATs raises various ethical and legal considerations that must be addressed to ensure that these technologies are developed, implemented, and used to respect the rights and dignity of individuals with disabilities. These considerations include privacy, autonomy, equity, and regulatory compliance.

Privacy and Confidentiality

With the integration of sensors, ATs may collect and store sensitive personal information, including health data, location, and usage patterns. Ensuring the data are securely stored and transmitted is paramount to protecting user privacy. Users must be fully informed about what data are being collected, how the data will be used, and who will have access to the data. Obtaining informed consent is essential to maintaining trust and transparency. Policies about data ownership need to be defined. The design and use of ATs must respect user privacy and ensure the confidentiality of personal and medical information. Users should be fully informed about the AT's benefits, risks, and limitations and provide consent before implementation. With specific video-based technologies

adopted, privacy-preserving solutions should be considered thoroughly.⁵² Caregivers and caring staff are susceptible to the possibility that technologies could be used for performance management.³⁴ How extensive data collected by monitoring technologies could constitute evidence of appropriate care delivery should be further investigated.

Autonomy and Independence

AT should be designed to empower users by enhancing their independence and enabling them to make their own decisions. Technologies should allow for customization to meet users' specific needs and preferences, providing them with control over how the device functions. Users should be provided with clear and comprehensive information about the available ATs, allowing them to make informed choices that best suit their needs.

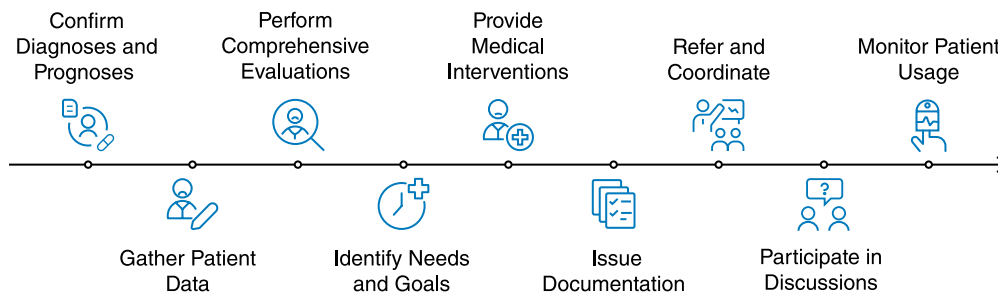
Equity and Fairness

The *UN Convention on the Rights of Persons with Disabilities* serves as a fundamental document guiding countries to adopt AT policies that promote accessibility and participation in society.³⁵ All individuals, regardless of their disability, should have equal access to ATs. Policies and practices should be in place to prevent discrimination and ensure equitable distribution. The development and use of ATs should respect the inherent dignity and human rights of individuals with disabilities. Efforts should be made to ensure that using ATs does not lead to stigmatization or marginalization of users. Efforts should be made to reduce the cost of ATs, making them affordable and accessible to individuals from all socioeconomic backgrounds. Government programs, insurance coverage, and other funding mechanisms should be in place to help offset the costs of assistive devices.

Legal and Regulatory Compliance

Compliance emphasizes that AT policies should be grounded in a human rights framework, ensuring that individuals with disabilities can access necessary technologies. It critiques the market-driven approach that views people with disabilities as consumers rather than rights holders.²⁵ ATs must comply with relevant laws and regulations, including safety, efficacy, and accessibility. Adherence to established standards (e.g., ISO, Americans With Disabilities Act of 1990) ensures that devices meet minimum performance and safety requirements. Ethical and legal considerations are integral to the responsible development and use of AT. By addressing issues related to privacy, autonomy, accessibility, equity, and regulatory compliance, stakeholders can ensure that ATs are both effective and respectful of the rights and dignity of individuals with disabilities. This approach enhances the impact of these technologies and promotes a more inclusive and equitable society. Finally, governments can leverage their power to enforce accessibility standards and stimulate the market for AT products.²⁵ Implementing comprehensive national AT plans ensures consistent access to AT across different regions and populations. These plans should involve multiple sectors, including health, education, and social welfare.²⁵ In recent years, sustainability has gained more attention in all fields. The Systemic Development Model, developed by the World Engagement Institute, addresses four interconnected pillars of sustainability: health, culture, economics, and politics. This model aims to improve understanding of personal, organizational, and institutional capacities.⁵ It encourages a holistic

Physician's Role in Assistive Technology Delivery



• Fig. 20.11 The physician's role in assistive technology delivery.

view beyond immediate health needs to include broader social, economic, and political dimensions, ensuring that the services provided are effective and sustainable in the long run. In addition, AT abandonment is still a potential problem. It has been reported that 29.3% of all devices were completely abandoned. Mobility aids were more frequently abandoned than other categories of devices, and abandonment rates were highest during the first year and after 5 years of use. Factors significantly related to abandonment are lack of consideration of user opinion in selection, easy device procurement, poor device performance, and change in user needs or priorities.⁵⁶

Physician's Role in Health Care Team in Assistive Technology Usage

In the field of AT, interdisciplinary collaboration is essential whether through interdisciplinary team models or transdisciplinary approaches. The delivery process for ATs could vary in different countries and societies. Not every type of AT requires assessment by medical personnel. Simple aids or devices, such as canes or knee braces, can be evaluated and delivered by trained personnel. However, for more complex devices or those with potential usage risks, it is essential for trained medical professionals rather than vendors to conduct the evaluation and prescription. Particularly when users have complex or severe conditions, a comprehensive assessment and recommendation by a physician may be necessary to ensure the effectiveness of the devices in the context of the illness. Physicians' pivotal roles involve confirming patients' information of diagnoses and prognoses, gathering the demographic and environmental data of patients, performing comprehensive evaluations of patients' body and organ functions, identifying needs and determining reasonable goals, providing appropriate medical interventions to optimize and prepare patients for the use of specialized ATs, issuing relevant documentation under regulations or policies, referring to and coordinating with appropriate teams for fitting and testing ATs, participating in interdisciplinary discussions, monitoring patient usage to identify any adverse reactions, and conducting research related to ATs. Physicians should also manage the coexisting medical problems to facilitate the use of ATs. For instance, a senior with osteoarthritis may limit the usage of particular ATs because of the pain. Adequate management and intervention will improve the adherence and acceptance of ATs. Proper training for the user and caregivers on how to use and maintain the technology is essential

for its effective utilization. Continuous follow-up and feedback from the user are necessary to identify any issues, provide ongoing support, and make necessary adjustments to the technology (Fig. 20.11).

Summary

Advances in AT are transforming the landscape of rehabilitation and daily living for individuals with disabilities. As we look to the future, the continued integration of emerging technologies, such as AI, robotics, and wearable devices, will drive further innovation and accessibility. By prioritizing personalization, affordability, and interdisciplinary collaboration, the future of ATs promises to enhance independence, productivity, and quality of life for individuals with disabilities.

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